

Agniveer Vayu Other than Science (Group Y) - 18 Jan 2023 - Memory Based Paper

English

Question 1

Select the correctly spelt word.

Options:

- A. Mathmetics
- B. Mathamatics
- C. Mathametics
- D. Mathematics

Answer: D

Solution:

The correct answer is '**Mathematics**'.

Key Points

- Out of the given options, only '**Mathematics**' is correctly spelt.
- '**Mathematics**' means the abstract science of number, quantity, and space, either as abstract concepts (*pure mathematics*), or as applied to other disciplines such as physics and engineering (*applied mathematics*).
- So, '**Mathematics**' is the correctly spelt word.

Hint

- Use the elimination method and try to eliminate the options.
 - For elimination, search for the easiest word which is known to you and eliminate the wrong options with the help of that word.
-

Question 2

Select the most appropriate synonym of the given word.

AGONY

Options:

- A. Pain
- B. Help
- C. Risk
- D. Blame

Answer: A

Solution:

The correct answer is 'pain'



Key Points

Let's look at the given words and their meaning.

- **Pain:** A state of suffering of the body or mind.
- **Help:** making it easier for someone to do something by offering one's services or resources
- **Risk:** exposure to danger, harm, or loss
- **Blame:** responsibility for a fault or wrong

Hence, the correct synonym of **agony** is **pain**

- **Example:** *The war veteran after getting injured on the battlefield exclaimed "Every moment I breathe is an agony"*



Additional Information

- The **synonyms** of **agony** are: anguish, misery, torture, woe, and distress.
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Question 3

Select the most appropriate **ANTONYM** of the given word.

LENIENT

Options:

- A. Generous
- B. Helpful
- C. Strict
- D. Defensive

Answer: C

Solution:

The correct answer is **option 3 i.e. Strict**.



Key Points

- **Let's look at the following points-**
- **Lenient**-means(of punishment or person in authority) more merciful or tolerant than expected. ((दंड या अधिकार में व्यक्ति) अपेक्षा से अधिक दयालु या सहनशील)
- **Strict**-(of a rule or discipline) demanding total obedience or observance; rigidly enforced. (एक नियम या अनुशासन का) पूर्ण आज्ञाकारिता या पालन की मांग करना; सख्ती से लागू किया गया)
- Hence, both options i.e. Lenient and Strict express opposite meanings.



Additional Information

Now, let's look at the following points-

- **Generous**-showing a readiness to give more of something, especially money, than is strictly necessary or expected. (कड़ाई से आवश्यक या अपेक्षा से अधिक, विशेष रूप से धन देने के लिए तत्परता दिखाना)
- **Helpful**-giving or ready to give help. (मदद देने के लिए तैयार)
- **Defensive**-used or intended to defend or protect. (बचाव या रक्षा के लिए इस्तेमाल या इरादा)

Thus, the correct option is **Strict** and therefore the answer.

Question 4

Complete the following sentence by choosing the correct option:

No sooner did we go out on the road _____ it began to rain heavily.

Options:

A. when

B. than

C. then

D. but

Answer: B

Solution:

The correct answer is '**than**'.



Key Points

- **No sooner** is used to show that one thing happens immediately after another thing. It is often used with the past perfect (also with past indefinite), and usually **followed by than**.
 - Structure: **No sooner + did + sub + V1 + extra info+ than + sub + V2 + extra info.**
 - *Ex: **No sooner** did they start their walk than it started to rain.*
- It's clear that in the blank given in the sentence, '**than**' should be filled.

The complete sentence is: *No sooner did we go out on the road **than** it began to rain heavily.*

*Therefore, the **correct answer is option 2).***

Question 5

Select the most appropriate option to complete the sentence.

He shouted _____ the top of his voice, but nobody heard him.

Options:

A. on

B. upon

C. with

D. at

Answer: D

Solution:

The correct answer is **option 4, i.e., at.**

- The sentence uses the blank after the verb and before the noun, which means that it must contain a preposition.
- Given the context, the blank should contain a preposition that means 'to' or 'at' the top of his voice, or the highest of his voice.
- The preposition '**at**' means denoting a particular point or level on a scale. viz, here, the top of his voice.
- None of the other prepositions can make the sentence meaningful.

Thus, the complete sentence is: *He shouted at the top of his voice, but nobody heard him.*

Question 6

Direction: Select a word or phrase from the given list to complete the sentence:

They needn't worry, _____?

Options:

- A. isn't it
- B. doesn't it
- C. don't it
- D. need they

Answer: D

Solution:

The correct answer is '**need they.**'



Key Points

- We need to identify the correct question tag.
- The given interrogative sentence consists of two parts.
- The first part is the statement "**They needn't worry**".
- The second part is the question tag.
- In such conditions, we need to use a **comma (,)** to separate the statement part from the question part.

- **Rules for question tag:**
 - The sentence and the question tag must be in the same tense.
 - If the sentence is positive, the question tag must be negative, and vice-versa.
 - Always use pronouns in the question tag.
 - The first part of the given sentence is a negative sentence.
 - Thus, a **positive question tag** should be used.
 - The helping verb in the given sentence is "**need**."
 - So, we have to use the positive question tag "**need they?**" for the given sentence.
- Therefore, the correct answer is **Option 4**.

Correct sentence: *They needn't worry, need they?*

Question 7

A sentence has been given with a blank to be filled with an appropriate word. Choose the correct alternative.

More than a decade has passed since this house was _____

Options:

- A. build
- B. built
- C. is build
- D. building

Answer: B

Solution:

Verb: Verbs are used to indicate the actions, processes, conditions, or states of beings of people or things. They play an integral role in the structure of a sentence. They constitute the root of the predicate, which, along with the subject (the "doer" of the verb's action), forms a full clause or sentence— we cannot have a sentence without a verb. It has three forms:.

For Example: Build (Base form) - Built (Simple Past) - Built (Past Participle)

Passive: A passive verb has a subject which is undergoing the action of the verb, rather than carrying it out, e.g. An apple was eaten. The opposite of active (He ate an apple). It should be noted that passive verbs are always in past participle form.

So, a suitable word is built.

Question 8

DIRECTIONS: Fill in the blank with correct preposition

Distribute the sweets equally _____ four children.

Options:

A. Between

B. In

C. Among

D. Through

Answer: C

Solution:

The correct answer is "**among**"



Key Points

- The correct **preposition** to use in this sentence is "**among**".
- "**Among**" is used when referring to **distribution or distribution of something among a group of people or things**.
 - **Example:** The teacher distributed the candy **among** the students.
- In this sentence, the sweets are being distributed among four children, so the correct **preposition** is "**among**."

Thus, *option 3* is the correct answer.

Correct sentence: *Distribute the sweets equally **among** four children.*



Additional Information

- "**Between**" is used when referring to a specific relationship or comparison between two things.
 - **Example:** The book is **between** the two chairs.
- "**In**" is used when referring to a specific location or container.
 - **Example:** The book is **in** the bag.
- "**Through**" is used when referring to movement or an action done in a specific direction.
 - **Example:** We walked **through** the park.



- "among" का अर्थ है किसी समूह के बीच या किसी समूह के बीच कुछ करने का है।
 - "between" का अर्थ है किसी विशिष्ट संबंध या तुलना के बीच किसी दो चीजों के बीच का है।
 - "in" का अर्थ है किसी विशिष्ट स्थान या कंटेनर के अंदर का है।
 - "through" का अर्थ है किसी दिशा या किसी काम को करने के दौरान का है।
-

Question 9

In the following question, out of the four alternatives, select the alternative which is the best substitute of the phrase.

That which cannot be believed

Options:

- A. Awesome
- B. Incredible
- C. Credible
- D. Ineffective

Answer: B

Solution:

Incredible =>that which cannot be believed

Awesome=>amazing, stunning

Credible =>acceptable, trustworthy

Ineffective=>something that is unsuccessful, useless.

So option 2 is the correct answer.

Question 10

Identify the segment in the sentence which contains a grammatical error.

Australia is one of the most prepossessing country in the Southern Hemisphere.

Options:

- A. Australia is one
- B. of the most prepossessing
- C. country in
- D. the Southern Hemisphere.

Answer: C

Solution:

The erroneous part is "**country in**"



Key Points

- "**countries**" will replace "**country**"
- Whenever any noun comes after the phrase "**one of the**" it must be plural.
- Look at the following examples:
 - Porsche is one my favourite **cars**.
 - Porsche is one my favourite car.
 - She is one of the best **girls** in the town
 - She is one of the best girl in the town.

Hence, the correct sentence is: *Australia is one of the most prepossessing **countries** in the Southern Hemisphere.*



Hinglish

- जब भी किस सेंटेंस में **one of the, most of the** का उपयोग होता है तो उसके बाद आने वाला नाम हमेशा प्लूरल होता है
- examples:
 - Tokyo is one of the large **cities** in Japan.
 - Most of the **students** do not follow the rules.
 - I am one of the most exquisite **boys** in my school
 - Most of the **girls** are ugly on the inside.

Question 11

Identify the segment in the sentence which contains a grammatical error.

Neither Elena nor they is going to win the lucrative basketball tournament.

Options:

- A. Neither Elena
- B. nor they
- C. is going to win
- D. the lucrative basketball tournament

Answer: C

Solution:

The erroneous part is "is going to win"



Key Points

- "are" will replace "is"
- If two subjects are joined by –Not only.....but also Neither.....nor Either.....or then, the verb always follows its nearest subject.
- Look at the following Examples:
 - Neither you nor I **am going** to see him.
 - Neither you nor she **is listening** to music.
 - Either Nicole or they **are performing** tonight.
 - Either Goretzka or Gnabry **is playing** today

Hence, the correct sentence is: *Neither Elena nor they **are** going to win the lucrative basketball tournament.*



Hinglish

- जब कभी भी दो expression, **Not only.....but also Neither.....nor Either.....or** से जोड़े जाते हैं तो verb हमेशा पास वाले noun के हिसाब से लगता है
 - जैसे इस सेंटेंस में **they** के हिसाब से लगा है:
 - Either Nicole or they **are performing** tonight.
 - Examples:
 - Neither you nor I **am going** to see him.
 - Neither you nor she **is listening** to music.
-

Question 12

Select the correct passive form of the given sentence.

He wrote a letter in verse form to his friend.

Options:

- A. A verse in letter form had been written by him to his friend.
- B. A letter in verse form was being written by him to his friend.
- C. A letter in verse form was written by him to his friend.
- D. A letter in verse form has been written by him to his friend.

Answer: C

Solution:

The passive voice of the given sentence is A letter in verse form was written by him to his friend.



Key Points

Simple Past Tense

- The **simple past** is a verb tense that is used to talk about things that happened or existed before now.

Active	Passive
Subject + V2+ object	Object+ was//were+V ₃ + by+ subject?

- The instructions given below should be followed while changing a **simple past sentence** to a **passive voice**.
 - Find the **subject and object** of the sentence and **exchange their places**; make changes in their **cases** as well if subject and object are **pronouns**.
 - Use the preposition '**by**' before the agent.
 - Use an appropriate **helping verb (was)** in the **passive form at the start of the sentence** according to the **tense of the active form**. (Simple past- was//were +v3)
 - Always use the **third form of the main verb (written)** in passive form.
 - At last line up the remaining part.

So, the final sentence is-A letter in verse form was written by him to his friend.

Question 13

Select the most appropriate indirect form of the given sentence.

He said, “Hurray! I have won the match.”

Options:

- A. He exclaimed with great sorrow that he had won the match.
- B. He exclaimed with great joy that he had won the match.
- C. He exclaimed with great joy that he would won the match.
- D. He exclaimed with great joy that he have won the match.

Answer: B

Solution:

The correct answer is '**He exclaimed with great joy that he had won the match**'.

Key Points

- The given sentence is an example of an **exclamatory sentence**.
 - The given sentence is in the form of **direct speech**. As per the question we **have to change it into indirect speech**.
 - Basic rules for changing or converting **direct speech into indirect speech**:
 - 'Comma' and 'inverted comma' are omitted in indirect speech and are often replaced by 'that' clauses to form 'that clauses'.
 - According to the reporting verb, changes are made in the direct sentence or the sentence in inverted commas.
 - The interjectional words or phrases are omitted, and the emotion is expressed through other words (**exclaim, cry out, pray, wish,**).
 - 'Said' will be changed into '**exclaimed with great joy**' in the **indirect speech**.
 - '**that**' conjunction will be added to account for the comma and double-inverted commas (**starting a direct speech**)
 - In an indirect sentence, the exclamatory sentence becomes an assertive sentence because exclamation (!) is replaced by a Full Stop (.) in indirect speech.
 - This is the direct and indirect speech rule for the **exclamatory sentence**.
 - **Example:**
 - *He said, “Hurrah! My friend has come”.* (direct speech)
 - *He exclaimed joyfully that his friend had come.* (indirect speech)
 - Therefore, as per the points mentioned above, we find that "**He exclaimed with great joy that he had won the match.**" is the correct indirect form of the given sentence.
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Question 14

Given here are four jumbled sentences. Pick the option that gives the correct order.

- A. So I had to try this place out.
- B. serves up the best Dutch apple pie in all of Utrecht.
- C. This is a magnificent restaurant right here which
- D. And I love Dutch apple pie.

Options:

- A. ABDC
- B. CBDA
- C. ABCD
- D. DCBA

Answer: B

Solution:

The correct answer is "CBDA"

Key Points

- The first sentence will be C because it introduces a restaurant
 - **This is a magnificent restaurant right here which**
- The second sentence will be B because it states the restaurant serves apple pie.
 - **serves up the best Dutch apple pie in all of Utrecht.**
- The third sentence will be D because it states that the doer loves apple pie.
 - **And I love Dutch apple pie.**
- The last sentence will be A because it states he wants to try out the restaurant
 - **So I had to try this place out.**

Hence, the correct passage is: *This is a magnificent restaurant right here which serves up the best Dutch apple pie in all of Utrecht. And I love Dutch apple pie. So I had to try this place out.*

Question 15

Given here are four sentences. The first sentence is in the correct order, while the rest of the sentences are jumbled. Pick the option that gives the correct order.

A. This is the Residenz and actually, I have just learned

B. which is very impressive given the fact the first structure

C. was built in 1395 and it continued being bigger and bigger and bigger.

D. that this is the largest city palace in all of Germany.

Options:

A. ABCD

B. ADBC

C. ACDB

D. ADCB

Answer: B

Solution:

The correct answer is "ADBC"



Key Points

- It is given in the question that the first sentence is in the correct order:
 - This is the Residenz and actually, I have just learned
- The next sentence will be D because it describes that the place is the largest palace in Germany
 - that this is **the largest city palace** in all of Germany.
- The next sentence will be B because it describes the impressiveness of the place
 - **which is very impressive** given the fact the first structure
- The last sentence will be C because it completes the previous sentence
 - **was built in 1395 and it continued being bigger and bigger and bigger.**

Hence, the correct passage: *A. This is the Residenz and actually, I have just learned D. that this is the largest city palace in all of Germany. B. which is very impressive given the fact the first structure C. was built in 1395 and it continued being bigger and bigger and bigger.*

Question 16

Directions: Read the following passage and answer the given questions:

Women's empowerment refers to the ability of women to have control over their own lives and make their own choices. This includes having equal access to education, employment opportunities, and the ability to participate in political and economic decision-making. To truly empower women, we must work to eliminate these attitudes and break down the systemic barriers that hold women back. This requires a multifaceted approach, including policies and programmes that promote gender equality and education and awareness campaigns that challenge stereotypes and biases. However, despite the progress that has been made in recent years, there remain indifferent attitudes towards women's empowerment. These individuals may not actively work to hold women back, but they also do not actively work to promote equality and empowerment. It is important to actively work to change these attitudes and promote a culture of equality and empowerment for all women.

What does "women's empowerment" refer to?

Options:

- A. Women's rights are not treated with the same respect and importance as men's rights.
- B. Women are denied basic rights, such as access to education.
- C. The ability of women to have control over their own lives.
- D. Many women are vulnerable to exploitation and abuse.

Answer: C

Solution:

The correct answer is "The ability of women to have control over their own lives."



Key Points

- Option '3' is correct.
 - From the passage: '....*Women's empowerment refers to the ability of women to have control over their own lives and make their own choices. This includes having equal access to education, employment opportunities, and the ability to participate in political and economic decision-making.....*'
 - This indicates that women's empowerment refers to **the ability of women to have control over their own lives and make decisions that affect them. It involves giving women equal opportunities and rights, and removing any barriers that prevent them from reaching their full potential.**
 - Thus, in this context, we can say that the correct answer is option "3".
- Options 1, 2, and 4 are incorrect because there isn't sufficient supportive information in the passage to consider them to be the right answers.

Hence, the correct answer is "*The ability of women to have control over their own lives.*"

Question 17

Directions: Read the following passage and answer the given questions:

Women's empowerment refers to the ability of women to have control over their own lives and make their own choices. This includes having equal access to education, employment opportunities, and the ability to participate in political and economic decision-making. To truly empower women, we must work to eliminate these attitudes and break down the systemic barriers that hold women back. This requires a multifaceted approach, including policies and programmes that promote gender equality and education and awareness campaigns that challenge stereotypes and biases. However, despite the progress that has been made in recent years, there remain indifferent attitudes towards women's empowerment. These individuals may not actively work to hold women back, but they also do not actively work to promote equality and empowerment. It is important to actively work to change these attitudes and promote a culture of equality and empowerment for all women.

Select the most appropriate ANTONYM of the given word.

"Indifferent"

Options:

- A. Apathetic
- B. Attentive
- C. Insensible
- D. Incurious

Answer: B

Solution:

The correct answer is “**Attentive.**”

Key Points

- **Indifferent** means showing or feeling no interest or concern in something. It suggests a lack of engagement or investment in a particular situation or thing. (उदासीन का अर्थ है किसी चीज़ में कोई दिलचस्पी या चिंता न दिखाना या महसूस करना। यह किसी विशेष स्थिति या चीज़ में जुड़ाव या निवेश की कमी का दर्शाता है।)
 - **For example,** "He was *indifferent* to her proposal and didn't seem to care either way."

Let's see the meanings of the given words

- **Apathetic:** This means showing or feeling no interest or concern in something. (इसका अर्थ है किसी चीज़ में कोई दिलचस्पी या चिंता न दिखाना या महसूस करना।)
 - **Example:** The crowd was *apathetic* towards the speeches given by the candidates.
- **Attentive:** This means paying close attention to something or someone. (इसका अर्थ है किसी चीज़ या किसी पर पूरा ध्यान देना।)
 - **Example:** The teacher was *attentive* to the needs of each student.
- **Insensible:** This means lacking feeling, emotion, or sensitivity. (भावना, संवेदनशीलता की कमी।)
 - **Example:** The soldier was *insensible* to the horrors of war.

- **Incurious:** This means showing a lack of interest or desire to know or learn more about something. (इसका अर्थ है किसी चीज़ के बारे में अधिक जानने या सीखने में रुचि या इच्छा की कमी।)
 - **Example:** *The child was **incurious** about the strange new toy.*

According to the meanings of the given words, the correct answer is **Option 2, i.e., Attentive.**

Question 18

Directions: Read the following passage and answer the given questions:

Women's empowerment refers to the ability of women to have control over their own lives and make their own choices. This includes having equal access to education, employment opportunities, and the ability to participate in political and economic decision-making. To truly empower women, we must work to eliminate these attitudes and break down the systemic barriers that hold women back. This requires a multifaceted approach, including policies and programmes that promote gender equality and education and awareness campaigns that challenge stereotypes and biases. However, despite the progress that has been made in recent years, there remain indifferent attitudes towards women's empowerment. These individuals may not actively work to hold women back, but they also do not actively work to promote equality and empowerment. It is important to actively work to change these attitudes and promote a culture of equality and empowerment for all women.

What is the importance of changing indifferent attitudes toward women's empowerment?

Options:

- A. Women are not valuable members of society and are not worthy of the same rights and opportunities as men.
- B. A culture of discrimination, where women are treated as second-class citizens.
- C. Women are subject to physical, sexual, and emotional abuse.
- D. It promotes a culture of equality and empowerment for all women.

Answer: D

Solution:

The correct answer is "**It promotes a culture of equality and empowerment for all women.**"

Key Points

- Option '4' is correct.
- From the passage '.... However, despite the progress that has been made in recent years, there still remains an indifferent attitude towards women's empowerment. **These individuals may not actively work to hold women back, but they also do not actively work to promote equality and empowerment. It is important to**

actively work to change these attitudes and promote a culture of equality and empowerment for all women.....?

- It helps to break down societal barriers and stereotypes that limit women's potential. When women are empowered, they are able to participate fully in society, contribute to the economy, and make decisions that affect their own lives and the lives of those around them. Thus, in this context, we can say that the correct answer is option "4".
- Options 1, 2, and 3 are incorrect because there isn't sufficient supportive information in the passage to consider them to be the right answers.

Hence, the correct answer is "It promotes a culture of equality and empowerment for all women."

Question 19

Directions: Read the following passage and answer the given questions:

Women's empowerment refers to the ability of women to have control over their own lives and make their own choices. This includes having equal access to education, employment opportunities, and the ability to participate in political and economic decision-making. To truly empower women, we must work to eliminate these attitudes and break down the systemic barriers that hold women back. This requires a multifaceted approach, including policies and programmes that promote gender equality and education and awareness campaigns that challenge stereotypes and biases. However, despite the progress that has been made in recent years, there remain indifferent attitudes towards women's empowerment. These individuals may not actively work to hold women back, but they also do not actively work to promote equality and empowerment. It is important to actively work to change these attitudes and promote a culture of equality and empowerment for all women.

What is required to truly empower women?

Options:

- A. An attitude shift and a shift in mindset towards gender equality.
- B. To eliminate typical attitudes and break down the systemic barriers that hold women back.
- C. Women have the same rights, abilities, and opportunities as men.
- D. Tackling the underlying causes of gender inequality in our societies.

Answer: B

Solution:

The correct answer is "**To eliminate typical attitudes and break down the systemic barriers that hold women back.**"

 **Key Points**

- Option '2' is correct.
 - From the passage' *However, despite the progress that has been made in recent years, **there still remains an indifferent attitude towards women's empowerment. These individuals may not actively work to hold women back, but they also do not actively work to promote equality and empowerment.***.....'
 - This indicates that women's empowerment is about breaking down societal and cultural barriers that prevent women from achieving equality and achieving their full potential. It involves creating a level playing field for women, so they have the same opportunities and rights as men. This can be achieved through various means, such as education, policies, and programs that promote gender equality and women's rights.
 - Thus, in this context, we can say that the correct answer is option "2".
- Options 1, 3, and 4 are incorrect because there isn't sufficient supportive information in the passage to consider them to be the right answers.

Hence, the correct answer is "To eliminate typical attitudes and break down the systemic barriers that hold women back."

Question 20

Select the verb form of the adjective.

Pacific

Options:

- A. pacify
- B. pacifying
- C. pacified
- D. pacification

Answer: A

Solution:

The correct answer is "pacify"

Key Points

- The correct verb form of the given word is "pacify".
- The adjective "**pacific**" means "**peaceful in character or intent.**" it is also act as a noun.
- The verb form of this is "**pacify**" which means "**quell the anger, agitation, or excitement of.**"
- "**pacified**" is a past participle and "**pacifying**" is a present participle.

- "pacification" is a noun that means "the action of bringing peace to a place or ending war in a place, often using military force"



Additional Information

Let's discuss the other options.

- **Verb:** It's an action word; it describes an action that someone does.
 - **Adverb:** it modifies a verb, adjective, or another adverb and typically describes how, when, where, or to what extent something happens.
 - **Interjection:** it's a word or phrase that expresses a strong feeling and is used to interrupt normal syntactical structure.
-

RAGA

Question 21

'MOM' a mission launched by ISRO stands for

Options:

- A. Moon Orbiter Mission
- B. Mars Orbiter Mission
- C. Mars Orbital Mission
- D. Mars Orbital Module

Answer: B

Solution:

The correct answer is Mars Orbiter Mission.



Key Points

- **Mars Orbiter Mission (MOM)**
 - The MOM mission is primarily a technological mission to study various aspects of Mars.
 - This mission was launched by ISRO on 5 November 2013.
 - This mission was launched from Satish Dhawan Space Centre, Sriharikota, located in Andhra Pradesh.
 - It was launched with the help of Polar Satellite Launch Vehicle (PSLV) C-25.
 - It is an indigenously-built space probe, India's first interplanetary mission, in the Martian orbit since September 24, 2014.

- India is the first country in the world to achieve Mars Orbit intersection in its 1st attempt.
- It is the cheapest interplanetary space mission in the world.

Additional Information

- **Indian Space Research Organization(ISRO)**
 - It is the national space agency of India.
 - It operates under the Department of Space that is directly overseen by the Prime Minister of India.
 - Headquarters: Bengaluru, Karanataka
 - Established on: 15 August 1969
 - Founded by: Vikram Sarabhai
 - Chairman: Sreedhara Panicker Somanath (also the Secretary of the Department of Space, Government of India).
-

Question 22

Which day is celebrated as Indian Air Force Day?

Options:

- A. 8 February
- B. 8 September
- C. 8 October
- D. 8 November

Answer: C

Solution:

The correct answer is '**8 October**'.



Key Points

Indian Air Force Day

- The Air Force was established on October 8, 1932, in undivided India under colonial rule.
- Therefore, Indian Air Force Day is celebrated every year on **8 October**.
- On this day a program is organized at Hindon Air Force Station in Ghaziabad.
- In this program, many veterans including army officers are involved and powerful aircraft are displayed in the sky.
- A parade of male and female pilots is held in Hindon on this day.
- In this ceremony, the Chief of the Air Staff felicitates the military personnel with medals.

Hence, the correct answer is **8 October**.

Additional Information

Indian Air Force

- The Indian Air Force is the fourth largest air force in the world.
- The Hindon Air Force Station located in Ghaziabad, Uttar Pradesh is the largest in Asia.
- Since the inception of the Indian Air Force, it has been following the path of its motto 'Nabh: Sprisham Deeptam'. It means 'To touch the sky with pride.' This motto of the Air Force has been taken from the 11th chapter of the Bhagavad Gita.
- The colors of the Indian Air Force are blue, sky blue, and white.

Important Points

Let's have a look at other options:

Date	Day
8 February	Safer Internet Day
8 September	International Literacy Day
8 November	World Radiography Day

Question 23

Droupadi Murmu before being the President of India was

Options:

- A. Governor
- B. Chief Minister
- C. Vice President
- D. Election Commissioner

Answer: A

Solution:

The correct answer is Governor.

Key Points

- **Droupadi Murmu**
 - She was the **Former governor of Jharkhand and tribal leader from Odisha**, Droupadi Murmu.
 - She was the first member of a scheduled tribe to be nominated for the position of **President of India**.
 - The **BJP-led NDA** officially nominated her for the **2022 presidential race**.
 - Droupadi Murmu is serving as **India's 15th president**.

- She triumphed over **Yashwant Sinha**, the joint candidate for the opposition.

Additional Information

- She is the first President born in Independent India and the **youngest person to hold the position**.
 - Before being chosen as president, she served in **a number of cabinet positions for the Government of Odisha** between 2000 and 2004.
 - She was the **ninth governor of Jharkhand** from 2015 to 2021.
-

Question 24

Where UNICEF Headquarter is situated?

Options:

- A. Geneva
- B. New York
- C. Paris
- D. Rome

Answer: B

Solution:

The correct answer is **New York**

Key Points

- UNICEF was formed initially to **provide emergency food and healthcare to children and mothers in countries** that had been **devastated by World War II**.
- In **1953** it became a **permanent part** of the **United Nations System**, and the words **international and emergency** were **dropped from the organization's name**, though it retained the original acronym, **UNICEF**.
- UNICEF **Headquarters** is located in **New York City** in the **United States**.

Important Points

- **UNICEF Foundation Day** is celebrated every year on **December 11** ever since the **United Nations General Assembly created UNICEF** on the same day back in **1946**.
- The program was initiated to provide supplies, and assistance and to improve the health, of children after World War II.
- In **October 1953**, the UNGA extended UNICEF's mandate indefinitely to assist vulnerable children.

Question 25

'Blue Revolution' is related to which of the following?

Options:

- A. Meat production
- B. Cereal production
- C. Fish production
- D. Pulses production

Answer: C

Solution:

The correct answer is **Fish Production**.

Blue Revolution is associated with **Fish Production** which is also known as **Nili Kranti**.

Key Points

- **Blue Revolution** is an **explosion in aquaculture**, and **specifically fish production** as a form of **self-sufficiency, nutrients, and economic growth for India**.
- The concept of **therapid increase in the production of fish and marine products** through **package programs** is called **Blue Revolution**.
- It was **launched** in India during the **7th Five year Plan (1985-1990)** when the central **government-sponsored the fish farmers development agency**.

Additional Information

- **Meat production** is related to **Red Revolution**.
 - **Cereal and Pulses production** is related to **Green Revolution**.
 - **Milk Production** is related to **White Revolution**.
-

Question 26

Which is the holy book of Sikhs?

Options:

- A. Ramayana
- B. Guru Granth Sahib
- C. Mahabharata
- D. the Bible

Answer: B

Solution:

The correct answer is Guru Granth Sahib.



Key Points

- **Guru Granth Sahib**
 - **The holy book of the Sikhs is known as the Guru Granth Sahib and it is considered as the eternal guru for all times after the tenth guru of the Sikhs, Guru Gobind Singh, declared that the lineage of living gurus will end with him. Hence, Statement 2 is correct.**
 - The original name of the Guru Granth Sahib was Adi Granth and it was Guru Gobind Singh who elevated it to the status of the eternal and immortal guru by naming the book, Guru Granth Sahib.
 - This holy book which comprises the sermons of the Gurus called the Gurbani is written in the Gurmukhi script, which incidentally is the official script for writing any Punjabi text in the state of Punjab.



Additional Information

- **Ramayana is one of the two major Sanskrit epics of ancient India and important text of Hinduism, the other being the Mahabharata.** The epic, traditionally ascribed to the Maharishi Valmiki, narrates the life of Rama, a legendary prince of Ayodhya city in the kingdom of Kosala.
- **The Bible is a collection of religious texts, writings, or scriptures sacred to Jews, Samaritans, Christians, Muslims, Rastafari's, and others.** It appears in the form of an anthology, a compilation of texts of a variety of forms that are all linked by the belief that they are collectively revelations of God.

All the SikhGuru's:

Sikh Guru's



SHRI GURU NANAK DEV JI

SHRI GURU ANGAD DEV JI

SHRI GURU AMAR DAS JI

SHRI GURU RAM DAS JI

SHRI GURU ARJAN DEV JI

SHRI GURU HARGOBIND SAHIB JI

SHRI GURU HARRAI JI

SHRI GURU HARKRISHAN SAHIB JI

SHRI GURU TEG BAHADUR JI

SHRI GURU GOBIND SINGH JI



Question 27

Which of the following is the capital of Greece?

Options:

- A. Athens
- B. Prague
- C. Bogota
- D. Budapest

Answer: A

Solution:

Athens is the capital of Greece.



Key Points

- The capital of the Czech Republic is Prague.
- The capital of Colombia is Bogota.
- The capital of Hungary is Budapest.



Additional Information

Greece:

- The first modern Olympics were held in Athens, Greece, in 1896.
 - The Olympic Games were held every four years.
 - Paris is to host the 2024 Olympics.
 - Athens is the capital of Greece.
-

Question 28

By selling an article for Rs. 480 a person lost 20%. For what should he sell it to make a profit of 20%?

Options:

A. Rs. 675

B. Rs. 720

C. Rs. 700

D. Rs. 750

Answer: B

Solution:

Given:

When SP Rs. 480 then loss = 20%

Formula Used:

If an article is sold at a profit of $x\%$, and the Cost Price is CP;

then the SP becomes:

$$SP = [(100 + x)/100] \times CP$$

Similarly, for a loss of $y\%$, the SP becomes:

$$SP = [(100 - y)/100] \times CP$$

Calculation:

Let the cost price of the article be X

According to the question,

$$\Rightarrow 80\% \text{ of } X = 480$$

$$\Rightarrow X = (480/80) \times 100 = 600$$

$$\text{Required profit \%} = 120\% \text{ of } 600 = \text{Rs. } 720$$

$\therefore$ For 20% profit he should sell on Rs. 720



Alternate Method

Given:

When SP Rs. 480 then loss = 20%

Formula used:

$$\text{New SP} = [(100 + \text{Gain\%}) / (100 - \text{Loss\%})] \times \text{Old SP}$$

Calculation:

$$\text{New SP} = [(100 + 20) / (100 - 20)] \times 480 = 120 \times 6 = \text{Rs. } 720$$

$\therefore$ For 20% profit he should sell on Rs. 720

Question 29

A and B alone can complete work in 9 days and 18 days respectively. They worked together; however 3 days before the completion of the work A left. In how many days was the work completed?

Options:

A. 5

B. 6

C. 7

D. 8

Answer: D

Solution:

Given:

A and B alone can complete work in 9 days and 18 days respectively.

Calculations:

Let the work be completed in x days.

According to the question,

A worked for $(x - 3)$ days, while B worked for x days.

$$\text{so } (x - 3)/9 + x/18 = 1$$

$$\Rightarrow (2x - 6 + x)/18 = 1$$

$$\Rightarrow 3x - 6 = 18$$

$$\Rightarrow 3x = 24$$

$$\Rightarrow x = 8$$

$\therefore$ the correct answer is 8 days.

Question 30

A thief is noticed by a policeman from a distance of 500 m. The thief starts running and the policeman chases him. The thief and the policeman run at the rate of 10 km and 12 km per hour respectively. What is the distance between them after 12 minutes?

Options:

A. 150m

B. 180m

C. 100m

D. 130m

Answer: C

Solution:**Given:**

Policeman notices a thief from a distance = 500m.

Policeman run at the rate of 12km/hr

And thief run at the rate of 10 km/hr

Distance covered time = 12minutes

Formula used:

Distance = speed x time

Calculation:

Relative speed = $(12 - 10)\text{km/hr} = 2 \text{ km/hr}$

Time = 12minutes = $12/60 \text{ hr} = 1/5\text{hr}$

So, Distance covered in 12minutes = Speed $\times$ Time = $2 \times (1/5) = 2/5\text{km} = 400 \text{ m.}$

$\Rightarrow$ Distance between the thief and policeman = $(500 - 400) = 100 \text{ m.}$

$\therefore$ After 12 minutes distance between thief and policeman is 100 m.

Question 31

A can do a piece of work in 6 days. 'B' destroy it in 4 days. A does work for 4 days, During the last 2 days of which B has been destroying A's work. How many more days must A work alone to complete the work?

Options:

A. 4

B. 7

C. 5.5

D. 5

Answer: D

Solution:

A completes the work in 6 days.

$\Rightarrow$ A's 1 day's work = $1/6$

$\Rightarrow$ A's 4 day's work = $4/6 = 2/3$

B destroys the work in 4 days.

$\Rightarrow$ B's 1 day's destruction = $1/4$

$\Rightarrow$ B's 2 day's destruction = $2/4 = 1/2$

According to the question,

Work done after 4 days = $(2/3) - (1/2) = 1/6$

$\Rightarrow$ Remaining work = $1 - 1/6 = 5/6$

A does $1/6$ work in = 1 day

$\Rightarrow$ A does $5/6$ work in = 5 days

$\therefore$ Number of more days A must work alone to complete the work = 5 days

Question 32

Two successive discounts of 10% and 20% are equivalent to a single discount of?

Options:

A. 28%

B. 27%

C. 25%

D. 30%

Answer: A

Solution:

Given:

The successive discount are 10% and 20%.

Formula used:

Successive discount = $X + Y - (XY)/100$

Where,

X = First discount

Y = Second discount

Calculation:

Successive discount = $X + Y - (XY)/100$

X = 10%, Y = 20%

$\Rightarrow 10 + 20 - 200/100$

$\Rightarrow 28\%$

$\therefore$ The single equivalent discount is 28%.



Alternate Method

Let the MP was Rs100

After first discount, price = $100 - 10 = \text{Rs. } 90$

After second discount, price = $90 - (90 \times 20)/100 = 90 - 18 = \text{Rs. } 72$

$\therefore$ Total single discount = $[(100 - 72)/100] \times 100 = 28\%$

Question 33

In an election, The winner gets 60% of the total votes and wins the election by 1200 votes then find the number of total voters in the election.

Options:

A. 6000

B. 8000

C. 10000

D. 12000

Answer: A

Solution:

Given:

The winner gets 60% of the total votes and wins the election by 1200 votes.

Calculation:

Let the total number of voters be x .

The number of votes that winner gets = 60% of x

$$\Rightarrow \text{Winner's vote} = 60 \times x/100$$

$$\Rightarrow \text{Winner's vote} = 3x/5$$

$$\text{Loser's vote} = (x - 3x/5)$$

$$\Rightarrow \text{Loser's vote} = 2x/5$$

$$\text{Margin of their votes} = \{(3x/5) - (2x/5)\}$$

$$\Rightarrow 1200 = x/5$$

$$\Rightarrow x = 6000$$

$\therefore$ The total number of voters in the election is 6000.

**Alternate Method**

As we know that winners gets 60% of total 100% votes then the loser will 40% votes and winner will win by 20% votes.

Let the number of voters be x

$$x \times 20/100 = 1200$$

$$\Rightarrow x \times 1/5 = 1200$$

$$\Rightarrow x = 6000$$

The total number of voters in the election is 6000

Question 34

Two dice are thrown simultaneously. Find the probability of getting a sum of more than 7

Options:

A. $\frac{7}{12}$

B. $\frac{1}{3}$

C. $\frac{5}{12}$

D. $\frac{19}{36}$

Answer: C

Solution:

Formula used:

Probability of occurrence of the event:

$$P(E) = \frac{n(E)}{n(S)}$$

Where,

$n(E)$ = Number of favorable outcome

$n(S)$ = Number of possible outcome

Dice: Small cube which has between 1 to 6 numbers on its sides.

Calculation:

When the dice are rolled, the possible outcomes are

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

$$n(S) = 6^2 = 36$$

Favourable outcomes are

(2,6), (3,5), (3,6), (4,4), (4,5), (4,6), (5,3), (5,4), (5,5), (5,6), (6,2), (6,3), (6,4), (6,5), (6,6)

$$\Rightarrow n(E) = 15$$

Using the above formula, the required probability

$$P = \frac{n(E)}{n(S)} = \frac{15}{36} = \frac{5}{12}$$

$\therefore$ Required probability is $\frac{5}{12}$.

Question 35

$$\tan 10^\circ \times \tan 20^\circ \times \tan 30^\circ \times \tan 45^\circ \times \tan 60^\circ \times \tan 70^\circ \times \tan 80^\circ = ?$$

Options:

- A. $\frac{1}{2}$
- B. 1
- C. $\frac{3}{2}$
- D. None of the above

Answer: B

Solution:

$$\Rightarrow \tan 10^\circ \times \tan 20^\circ \times \tan 30^\circ \times \tan 45^\circ \times \tan 60^\circ \times \tan 70^\circ \times \tan 80^\circ$$

$$\Rightarrow \tan(90 - 80)^\circ \times \tan(90 - 70)^\circ \times \tan(90 - 60)^\circ \times \tan 45^\circ \times \tan 60^\circ \times \tan 70^\circ \times \tan 80^\circ$$

$$\Rightarrow \cot 80^\circ \times \cot 70^\circ \times \cot 60^\circ \times \tan 45^\circ \times \tan 60^\circ \times \tan 70^\circ \times \tan 80^\circ$$

$$\Rightarrow \tan 45^\circ = 1$$

Question 36

Find the area of equilateral triangle, if the side of an equilateral triangle is $16\sqrt{3}$ cm.

Options:

- A. $134\sqrt{3} \text{ cm}^2$
- B. $192\sqrt{3} \text{ cm}^2$
- C. $144\sqrt{3} \text{ cm}^2$

D. $156\sqrt{3}\text{ cm}^2$

Answer: B

Solution:

Given

Side of an equilateral triangle = $16\sqrt{3}\text{ cm}$

Formula used

Area of an equilateral triangle = $(\sqrt{3}/4) \times (\text{side})^2$

Calculation

Area of an equilateral triangle = $(\sqrt{3}/4) \times (16\sqrt{3})^2$

$$\Rightarrow (\sqrt{3}/4) \times (16\sqrt{3}) \times (16\sqrt{3})$$

$$\Rightarrow \sqrt{3} \times 4\sqrt{3} \times 16\sqrt{3}$$

$$\Rightarrow 48 \times 4 \times \sqrt{3}$$

$$\Rightarrow 192\sqrt{3}\text{ cm}^2$$

$\therefore$ Area of an equilateral triangle is $192\sqrt{3}\text{ cm}^2$

Question 37

If $\sin x + \operatorname{cosec} x = 2$, then $\sin^2 x + \operatorname{cosec}^2 x =$ _____ .

Options:

A. 3

B. 2

C. 4

D. 1

Answer: B

Solution:

Formula used:

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$\sin\theta \times \operatorname{cosec}\theta = 1$$

Calculation:

$$\sin x + \operatorname{cosec} x = 2$$

$$\Rightarrow (\sin x + \operatorname{cosec} x)^2 = 2^2$$

$$\Rightarrow \sin^2 x + \operatorname{cosec}^2 x + 2(\sin x)(\operatorname{cosec} x) = 4$$

$$\Rightarrow \sin^2 x + \operatorname{cosec}^2 x + 2 = 4 \quad (\because \sin\theta \times \operatorname{cosec}\theta = 1)$$

$$\therefore \sin^2 x + \operatorname{cosec}^2 x = 2$$

Question 38

If DRIVER = 12, PEDESTRIAN = 20, ACCIDENT = 16, then CAR will be-

Options:

- A. 3
- B. 6
- C. 8
- D. 10

Answer: B

Solution:

The logic followed here is,

$$\text{DRIVER} = 6 \times 2 = 12$$

$$\Rightarrow \text{Number of letters} \times 2$$

$$\text{PEDESTRIAN} = 10 \times 2 = 20$$

$$\text{ACCIDENT} = 8 \times 2 = 16$$

Similarly,

$$\mathbf{CAR} = 3 \times 2 = \mathbf{6}$$

Hence, the correct answer is "6".

Question 39

Find the next number of the series.

45, 54, 65, 78, 93, ?

Options:

A. 112

B. 111

C. 110

D. 113

Answer: C

Solution:

The logic followed here is:



Hence, the correct answer is "option (3)".

Question 40

In a row of 28 students, Jamuna was 12th from the extreme left end and Ganga was 19th from the extreme right end. If Sarasvati was seated

exactly between them, then what was Sarasvati's position from the right end? Assume that the students were seated facing north.

Options:

A. 17th

B. 16th

C. 20th

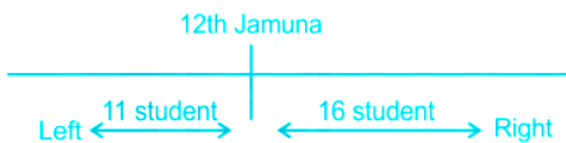
D. 18th

Answer: D

Solution:

1. In a row of 28 students, Jamuna was 12th from the extreme left end.

- This implies that Jamuna was 17th from the extreme right end.



2. Ganga was 19th from the extreme right end.

3. If Sarasvati was seated exactly between them.



Clearly, Sarasvati is 18th from the right end.

Hence, '18th' is the correct answer.

Question 41

How is your maternal uncle's only sister's husband's brother related to you ?

Options:

- A. Paternal Uncle
- B. Maternal Uncle
- C. Brother
- D. Grandfather

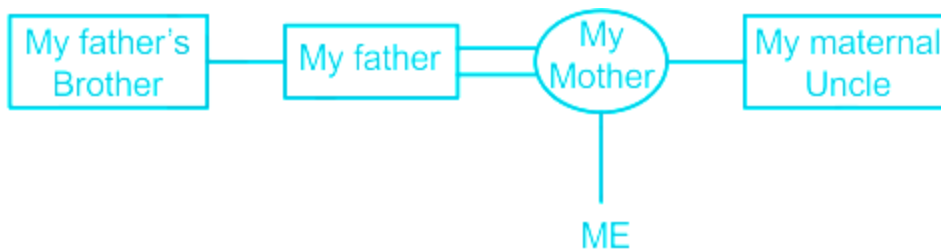
Answer: A

Solution:

In this question, we use the table given below:

Symbol in Diagram	Meaning
○	Female
□	Male
══	Married Couple
—	Siblings
	Difference of A Generation

According to the given information, the family tree is given below:



From the above figure, we can see that **my maternal uncle's only sister is my mother and my mother's husband is my father and my father's brother is my paternal uncle.**

Hence, "**Option 1**" is the correct answer.

Question 42

If M denotes '-', N denotes '÷', O denotes '×', and P denotes '+', then what will come in place of '?' in the following equation?

$$324 \text{ P } 35 \text{ M } 184 \text{ M } 17 \text{ O } 3 \text{ P } 39 = ?$$

Options:

A. 163

B. 171

C. 159

D. 153

Answer: A

Solution:

B	Brackets in order (), {}, []	ब्रैकेट (), {}, [] क्रम में
O	of	का
D	Division (÷)	विभाजन (÷)
M	Multiplication (×)	गुणा (×)
A	Addition (+)	जोड़ (+)
S	Subtraction (−)	घटाव (−)

Symbol	Meaning
M	-
O	×
P	+

Given equation: $324 \text{ P } 35 \text{ M } 184 \text{ M } 17 \text{ O } 3 \text{ P } 39$

By putting values and applying BODMAS rule we get,

$$324 + 35 - 184 - 17 \times 3 + 39$$

$$\Rightarrow 324 + 35 - 184 - 51 + 39$$

$$\Rightarrow 398 - 184 - 51$$

Hence, "Option 1" is the correct answer.



Additional Information

There is 2 M is given in the above equation.

Question 43

Six people, A, B, C, D, E, and F are seated in a row facing north. Only two people sit to the right of C. Only two people sit between B and C. F sits fourth to the right of B. E is an immediate neighbour of A and B. What is the position of D?

Options:

- A. Third to the left of A
- B. Third to the right of E
- C. Third to the left of E
- D. Third to the right of A

Answer: D

Solution:

According to the given information;

Only two people sit to the right of C-



Only two people sit between B and C-



F sits fourth to the right of B-



E is an immediate neighbour of A and B-



According to the final seating arrangement, D sits third to the right of A.

Hence, **option (4)** is the correct answer.

Question 44

Find the next number of the series.

0, 3, 8, 15, ?

Options:

- A. 24
- B. 23
- C. 25
- D. 22

Answer: A

Solution:

The logic followed here is:



Hence, the correct answer is **"option (1)"**.

Question 45

Select the pair from the alternatives that is similar to the pair given in the question.

Tokyo : Japan : : Vienna : ?

Options:

- A. Australia
- B. Austria
- C. China
- D. Cambodia

Answer: B

Solution:

The logic followed here is:

Tokyo is the capital of Japan.

Similarly,

Vienna is the capital of Austria.

Hence, the correct answer is "option (2)".

Question 46

Find odd one out.

Tehran, Bangladesh, Pakistan, Myanmar

Options:

- A. Tehran

B. Bangladesh

C. Pakistan

D. Myanmar

Answer: A

Solution:

The logic followed here is:

All are the neighbouring country of India except Tehran.

So, the odd one out is "**Tehran**".

Hence, the correct answer is "**option (1)**".

Question 47

If ROAD is written as "WTFI" in certain code language then what will be the code of BEAT ?

Options:

A. GJFY

B. GFJY

C. GFYJ

D. YFJG

Answer: A

Solution:

The logic followed here is:

R	O	A	D
+5↓	+5↓	+5↓	+5↓
W	T	F	I

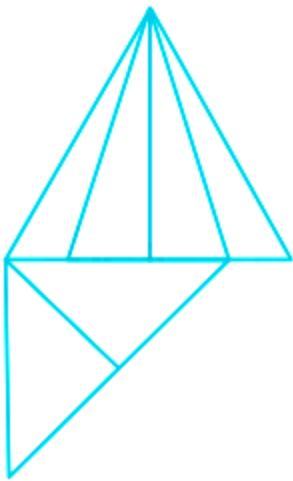
Similarly,



Hence, the correct answer is "option (1)".

Question 48

Find how many triangles are there in the given figure.



Options:

A. 12

B. 13

C. 14

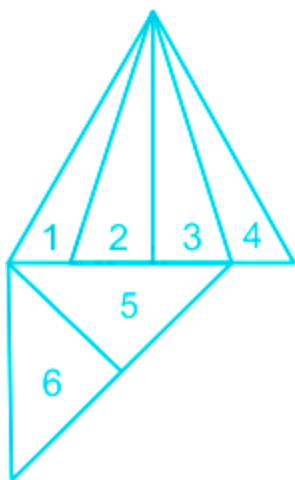
D. 11

Answer: B

Solution:

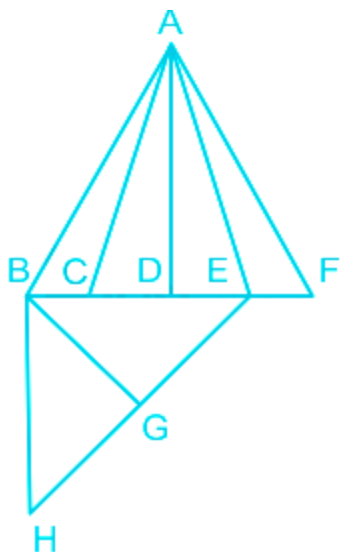
Here, we have an image and count total number of triangles.

First we will count small triangles as below:



So, we have a 6 triangles.

Next, we will combine small triangles and count them as below:



So, here the triangles are: BAD, BAE, BAF, CAE, CAF, DAF, HBE - a total of 7 triangles.

Thus, in all we have $6 + 7 = 13$ triangles.

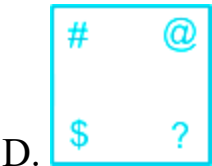
Hence, we have 13 triangles in all.

Question 49

Select the figure from the options that can replace the question mark (?) in the following figure series.



Options:



Answer: C

Solution:

The images in the figure are shifting its position by rotating in:

- For '@'- It is shifting in the anti-clockwise direction.



- For '\$'- It is shifting in the anti-clockwise direction.



- For '#' - It is shifting in the anti-clockwise direction.

\$?	? ← #	# @	@ \$	\$?
@ #	\$ @	? \$	# → ?	@ #

- For '?' - It is shifting in the anti-clockwise direction.

\$ ← ?	? #	# @	@ \$	\$?
@ #	\$ @	? → \$	# ?	@ #

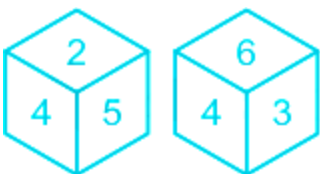
- For 'LINE' () - It is not shifting and are in the decreasing order.

\$?	? #	# @	@ \$	\$?
@ #	\$ @	? \$	# ?	@ #

Hence, "Option - (3)" is the correct answer.

Question 50

Two different positions of the same dice are shown, the six faces of which are numbered from 1 to 6. Select the number that will be on the face opposite to the one showing '4'.



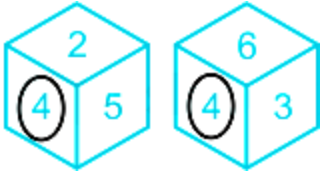
Options:

- A. 2
- B. 5
- C. 1
- D. 3

Answer: C

Solution:

The logic followed here is:



Here, we have two dices and face number 4 is common in both of them.

Also, both the dice are arranged in such a way that the common faces are at the same place.

So, we can see that, all the numbers except number 1 is not seen here in the image.

Hence, the number 1 will be the face opposite to the face 4.
